2007 F=ma Exam: Problem 19

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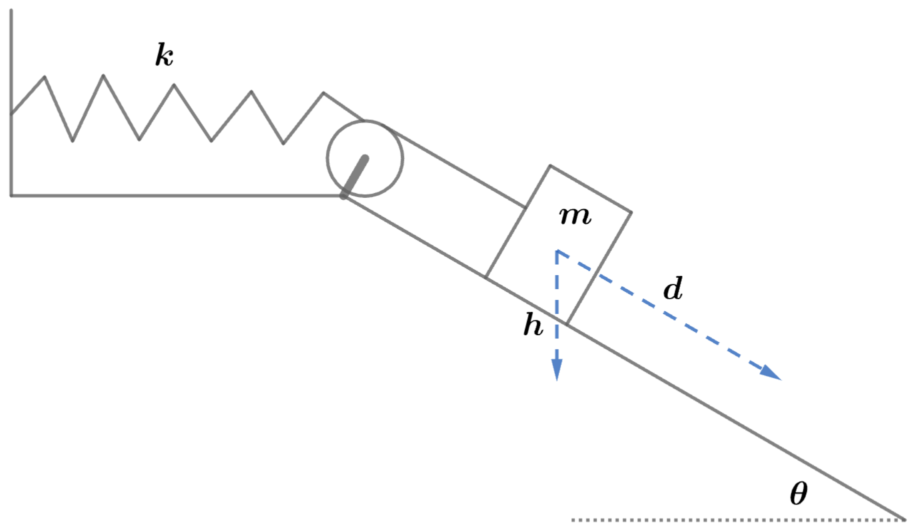

We have a non-Hookian spring with force $F(x) = -kx^2$ so its corresponding potential energy is

$$U(x) = -\int_0^x F(s)\,ds = \int_0^x ks^2\,dx = \frac{1}{3}kx^3$$

Once the mass is released, it will momentarily be at rest when the loss in gravitational potential energy balances the gain in spring potential energy:

$$mgh = \frac{1}{3}kd^3$$

Since $\theta = 30^\circ$ and $h = d\sin\theta = d/2$,

$$\frac{mgd}{2} = \frac{kd^3}{3}$$

$$d = \sqrt{\frac{3mg}{2k}}$$

so the answer is $\boxed{\text{A}}$.